Population or sample

1 True or false: In a sample survey, information is obtained from the entire population.

2 State whether the following is an instance of a sample or a census:

a A random selection of some people at a mall.

b A stock take of all the goods in store.

c A crash test of new cars just manufactured by a factory.

d Asking all the teachers at your school whether they approve of a new class timetable.

e An election to decide the premier of Queensland.

f Asking a random selection of students in your class whether they approve of the teacher.

g A taste test of a large batch of cookies you have just baked.

h A body scan of randomly selected passengers at Melbourne International Airport.

3 State whether a sample or a census would be more suitable for determining the following:

a The number of ambulances in Australia.

b The number of people watching Stranger Things.

c The average number of letters in the surnames of teachers at your school.

d The number of smokers in South Australia.

e The average commuting time of teachers at your school.

f The average height of men in Spain.

g The number of traffic lights in the suburb you live in.

h The most popular candidate in an election.

Sampling bias and misleading statements

4 Sean wants to know what 8th graders think of their English class, so he polls 70 random 8th graders. State whether the sample chosen is biased or fair.

5 A school principal wants to estimate the number of students who ride a bicycle to school. State whether the following samples would avoid bias:

a All students who are in the school band.

b Eight students in the hallway.

c Ten students from each grade, chosen at random.

d 130 students during the lunch periods.

6 State whether the following scenarios use[illegible]ed sampling methods:

- **a** A community nurse wants to know the average height of all 7th graders that attend the school where she visits, so she measures the height of all the basketball players.
- **b** A city councilman asks members of the ice hockey team if they would prefer a new skateboard park or a new ice-skating rink to be built as the new building project.
- **c** The lifeguard of a water park wants to determine which water rides are enjoyed the most so he asks every tenth person who leaves the park to list their three favourite rides.

7 Explain why the following samples are biased:

- **a** Hannah is surveying customers at a shopping precinct. She wants to know which stores customers shop at the most. She walks around an entertainment store and chooses 30 customers from the store for the survey.
- **b** A TV station wants to know what the most popular type of music is, so they ask listeners to contact them and vote for their favourite type of music.

8 An online company asks all its customers to complete a survey and 58% respond. In the survey, there was a question which asked “How happy are you with our service?”.
Respondents could tick either “Very Happy” or “Somewhat Happy”.
In their annual newsletter, the company wrote an article reporting “100% of customers are happy with our service”.
Explain why this claim is misleading.

9 Lachlan asks 120 Year 12 students at his school how much time they spend on homework per night. 78 Year 12 students say they do more than 3 hours. At a meeting of the student council, Lachlan reports "65% of students at this school do too much homework."
Determine whether the following statements explain why Lachlan's report is misleading:

- **a** The survey does not represent the population of the school.
- **b** The question should have been multiple choice.
- **c** The question was biased.
- **d** The sample size was too small.

10 In the polls for an upcoming election, various polls are analysed to see who is likely to win the election. Most polls put Kodos ahead of Kang by between 10% and 12%. A few polls have Kang ahead of Kodos by a few percentage points.
Kang tells his followers that they are sure to win the election based on the latest polls.
Explain why this statement misrepresents the statistics.

Types of sampling

11 For each item, detemine the type of sampling method used for these following events:

a Drawing out the winning ticket number in a lottery.

b Choosing every 50th person on the class roll to take part in a survey.

c Choosing 5% of the of the students in each year for Years 7-12.

d Standing at the drive-through at a fast food chain, asking customers if they will participate in a survey.

e Asking people questions until a particular number of respondents in a category is filled.

12 A television station wants to estimate the number of viewers it had for a new show. They know that the country's population is $13\,620\,000$. When they randomly selected 5000 people and asked them if they had watched the show, they found that 400 of them said 'Yes'. Estimate the number of people who watched the show in the entire population.

13 Out of 2160 students in a school, 216 were chosen at random and asked their favourite colour out of red, blue and yellow with 99 choosing red, 63 blue and 54 yellow.

a One in every how many students at the school was sampled?

b Estimate the total number of students in the whole school who prefer the colour:

i Red ii Blue iii Yellow

14 A factory produces 1980 laptops every day. How many laptops are tested daily if the factory tests a systematic sample of 1 out of every:

a 11 laptops. b 90 laptops.

15 A factory produces 3432 iPhones every day. One in every how many iPhones needs to be tested, if the factory is to test a systematic sample of size:

a 13 iPhones per day. b 88 iPhones per day.

16 In a group of 2160 students, 960 are male and 1200 are female. A stratified sample of 18 is to be selected from the group based on gender.

a How many males should be selected? b How many females should be selected?

17 A group of people is divided into four tea[redacted]lue, Red, Green and Yellow. The table shows the number of people in each team:

Team	Number of people
Blue	150
Red	390
Green	270
Yellow	300

a How many people are there combined in all of the teams?

b If a stratified sample of 1 in 30 is to be taken from the group, state the size of the sample.

c For the sample to be stratified, find the number of people that should be chosen from each team:

i Blue team **ii** Red team

iii Green team **iv** Yellow team

18 Four lucky people from a group of 215 each stand to win an iPad. Every contestant is given a different number between 1 and 215, and the winners are selected by generating a random number uniformly between 0 and 1. To ensure there is an equal chance of each contestant winning, the number is multiplied by 215 and then rounded up.
In this case, the numbers generated were:

$$0.152,\ 0.534,\ 0.352,\ 0.795$$

Convert the generated numbers into the numbers of the four winning contestants.

19 A manager wants to randomly select products on an assembly line to test their quality. She generates a random number between 2 and 15, which tells her how many products to pass before picking up the next one. She then generates another random number and so on.
The first product she picks up is the first one on the assembly line. She then generates the following numbers:

$$10,\ 11,\ 7,\ 14,\ 4$$

a How many products did she test in total?

b How many products did she pass before picking up the second product?

c How many products did she pass between the third and fourth tests?

d How many products were in front of the third one she tested?

20 In a study of asthma sufferers, a group of [redacted] are asked to identify which category they fit into:

- A - developed asthma from ages 0 to 10
- B - developed asthma in their teens
- C - developed asthma in their twenties
- D - developed asthma after the age of 30

Researchers then generated random values between 0 and 1 to decide which groups to choose participants from. These random values are shown in the given table:

0.750	0.574	0.145	0.154	0.564
0.752	0.580	0.573	0.427	0.197
0.144	0.634	0.399	0.295	0.787
0.971	0.643	0.313	0.169	0.979

a If the number was less than 0.25, the participants were chosen from Category A. How many were chosen from Category A?

b If the numbers were greater than 0.75, the participants were chosen from Category D. How many were chosen from Category D?

21 Prior to an election, a news editor wanted to view the opinions of randomly selected people. Some two-digit numbers were randomly generated, as shown in the table, and starting with 37, the editor moved 3 to the right, 1 down (so that the next number chosen would be 57). The numbers chosen were the ages of the people she would survey.

37	49	19	72	38	33	80	83	90	23	78	21	29	72	40
85	79	84	57	46	49	53	55	51	36	66	61	86	66	41
74	71	40	49	42	17	50	68	27	15	47	70	47	63	32
37	33	84	34	35	51	50	87	65	47	38	78	80	39	60
23	82	64	22	21	76	38	67	43	75	39	76	72	48	33

a List all the random numbers the editor chose.

b Find the range for the ages of the people who will be surveyed.

22 The following table shows the gender of f[redacted]ear 12 students at a particular school:

Students 1 - 10	M, F, M, F, F, M, M, M, F, M
Students 11 - 20	M, F, M, M, F, M, F, M, F, F
Students 21 - 30	F, F, M, M, F, M, M, F, F, F
Students 31 - 40	M, F, M, F, F, F, M, M, M, F
Students 41 - 50	F, M, F, M, M, F, M, F, M, F

a Calculate the proportion of females in the sample.

b What proportion of the first 5 students were female?

c What proportion of the first 10 students were female?

d What proportion of the first 20 students were female?

e In a systematic sample, every second student is chosen, in the order that they appear, from the first 20 students. How many males will be chosen in the sample?

f In a systematic sample, every third student is chosen, in the order that they appear, from the first 40 students. How many females will be chosen in the sample?

g The school has a population of 440 students. If the proportion of males and females in the sample is indicative of the whole school, how many female students are there in the school?